From preferences to rewards: A brief introduction

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Iliad Intensive

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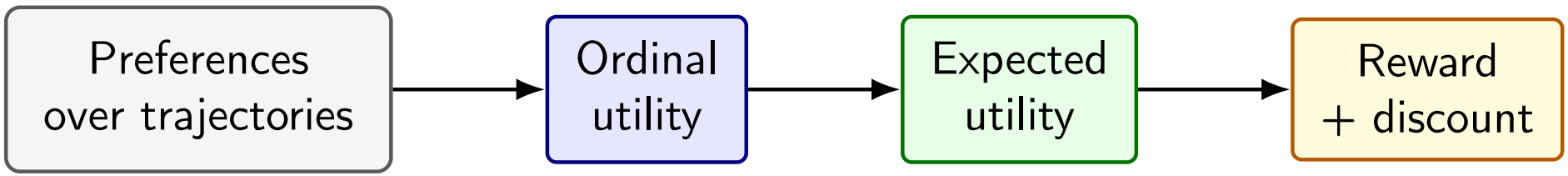
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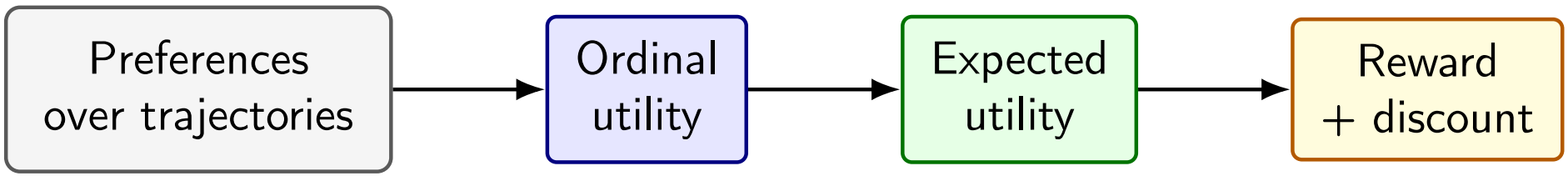
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- What are goals? Hard question.
- **Goal of today**: understand what are preferences.

The Route



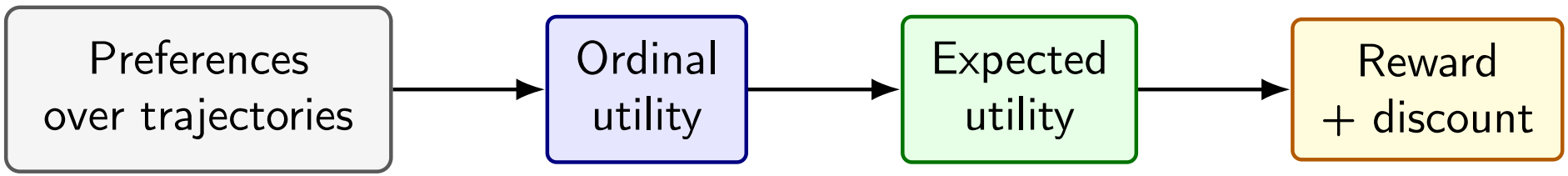
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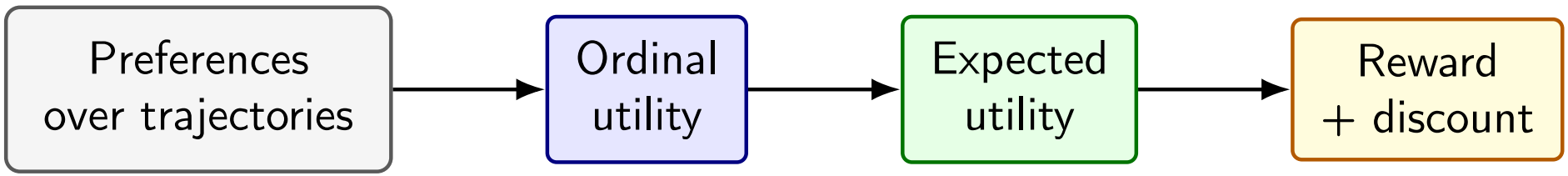
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- *Outcome 1*: Understand the assumptions of each step.
- *Outcome 2*: Recognise the difference between preference, utility, and reward.

Basic setting

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- The full finite trajectory space is

$$\mathcal{H}^* = \bigcup_{n=0}^{\infty} (\mathcal{O} \times \mathcal{A})^n.$$

Preferences

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- At this point, $\succcurlyeq$ need not be complete, transitive, numerical, or 'reward-like'.

Three Kinds of Trouble

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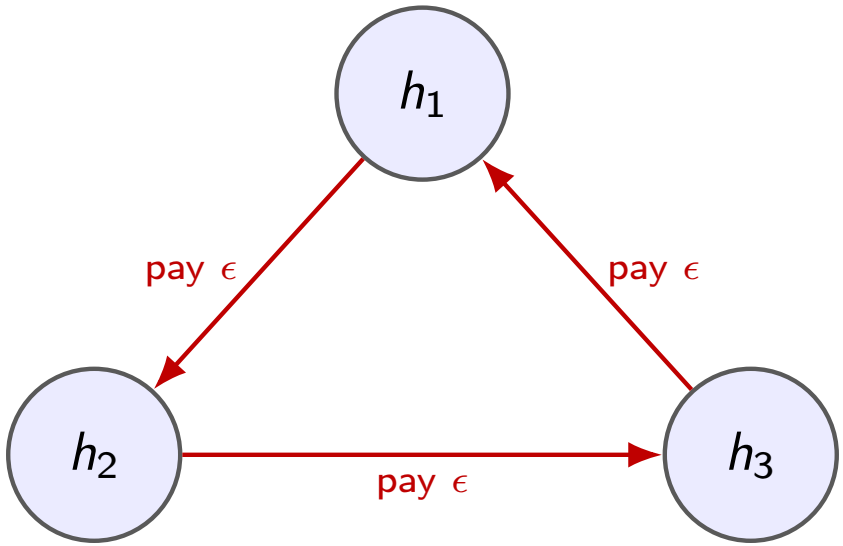
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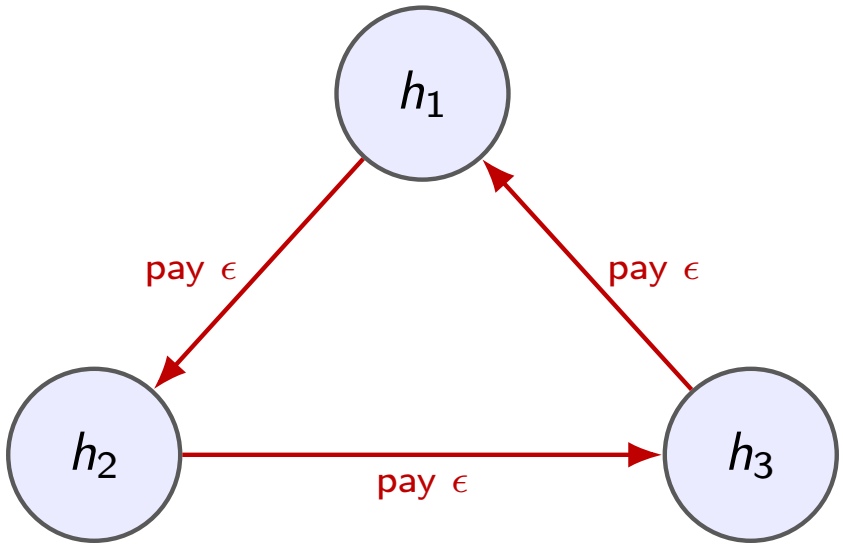
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  - E.g. **dominance**: another policy yields prefered trajectories at least as preferred in every state of the world, and strictly so in at least one.
- **Vulnerability:** individually acceptable trades can combine into guaranteed loss.
  - E.g. **Money pumps**: circular preferences ($h_1 \succ h_2 \succ h_3 \succ h_1$) lead to indefinite loss.

Money Pump Picture



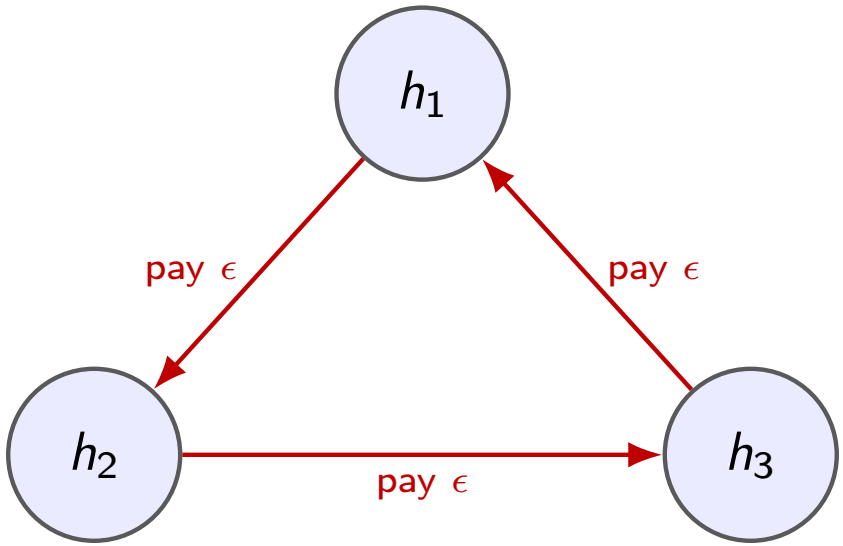
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- It returns to the same trajectory with less money.
- Parallel with thermodynamics: the second law of thermodynamics can be read as stating that *'nature cannot be money-pumped'*.

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These results are related (one can lead to the other) but not identical: form, vulnerability, and survival are different claims.

Axiom 1: Completeness

Completeness says every pair can be compared:

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- If it fails, some trajectories are genuinely incomparable.

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- The classic failure is a cycle: money pumps.

$$h_1 \succ h_2, \qquad h_2 \succ h_3, \qquad h_3 \succ h_1.$$

- Similar to Helmholtz-Hodge decomposition: transitivity is equivalent to requiring no rotational component.

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- Only the ranking matters:

$$u, \quad 2u + 17, \quad \exp(u)$$

all represent the same deterministic preferences.

Beyond Ordinal Utility

Ordinal utility ranks outcomes, but give a structure to mixtures of them.

- Example:

$$x \succ y \succ z.$$

How should we rank y with respect to $\frac{1}{2}x + \frac{1}{2}z$? Should they be equivalent?

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Ordinal rank doesn't say how to rank mixtures, which can arise naturally out of uncertainty.

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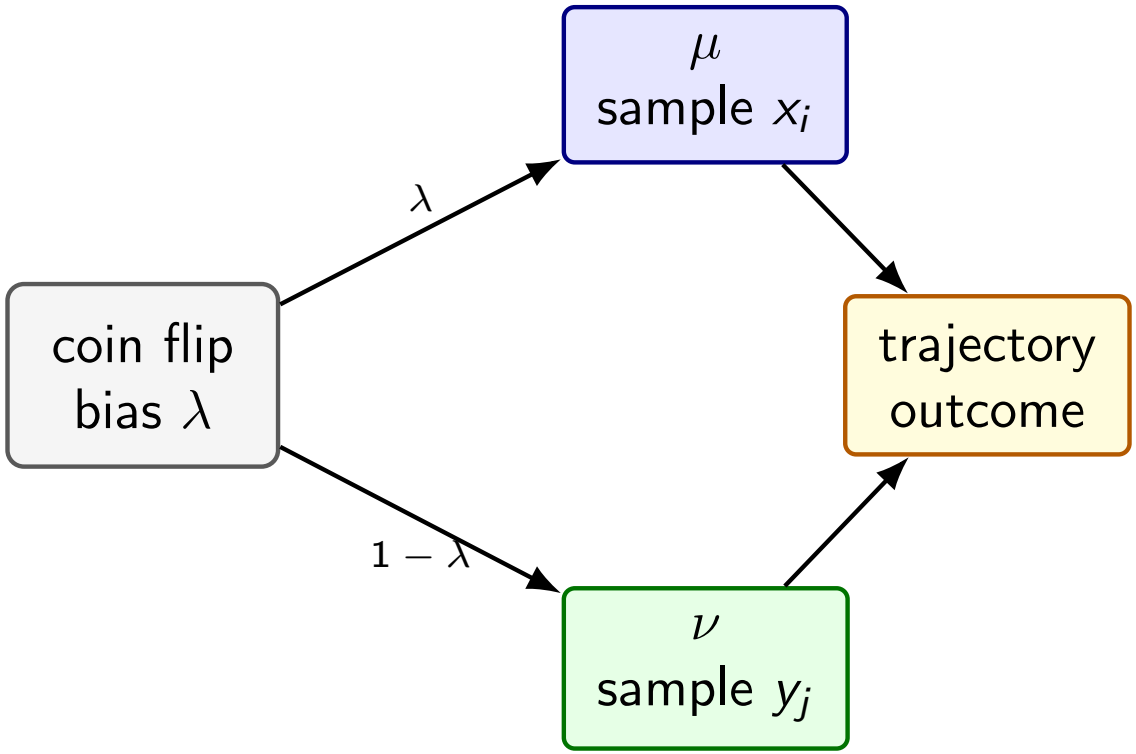
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- A sure trajectory x is the degenerate lottery δ_x.
- We can mix lotteries:

$$\lambda\mu + (1-\lambda)\nu.$$

Lottery Picture



- vNM asks when preferences over these objects are expectations.

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Result: The first two axioms give ranking structure based on a cardinal utility; the last two axioms make the utility to be an expected value.

Axiom 3:Continuity

- If $\mu \succcurlyeq \nu \succcurlyeq \xi$, continuity says there always exists some value $\lambda \in [0, 1]$ such that

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Classic counter-examples are lexicographic preferences, in which events can be infinitely different from each other.

Example: μ is eating chocolate, ν is eating potatos, and ξ is accepting genocide.

Axiom 4:Independence

- If $\mu \succcurlyeq \nu$, then mixing both with the same background lottery ξ should not reverse the comparison:

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When it fails, common consequences do not psychologically or behaviorally cancel.

Example (Allais paradox): Most people would say

- Getting \$100 for sure *is better than* a coin flip for \$250.
- A 1-in-100 shot at \$100 *is worst than* a 1-in-200 shota at \$250.

These are the same choice, just scaled down.

Theorem 1: The von Neumann-Morgenstern representation

Assumptions: a finite set of options $X \subseteq \mathcal{H}^*$, the four axioms hold.

- **Result 1 (existence):** there exists $u : X \to \mathbb{R}$ such that

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- **Result 2 (uniqueness):** u is unique up to positive affine transformations:

$$u'(x) = a + bu(x), \qquad b > 0.$$

Ordinal vs expected utility

Ordinal utility

- Any set of outcomes
- Preserves ranking
- Indiferent to non-decreasing transformation
- Cannot display ‘meaningful gaps’

vNM utility

- Lotteries (probabilistic mixtures)
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vNM utility futher refines ordinal utility (it is not just a relabeling).

Beyond expected utility: Temporal structure

- vNM utility assigns a single scalar to a whole trajectory.

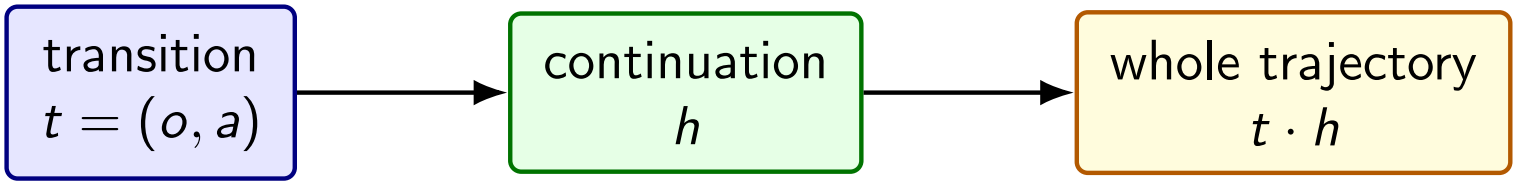
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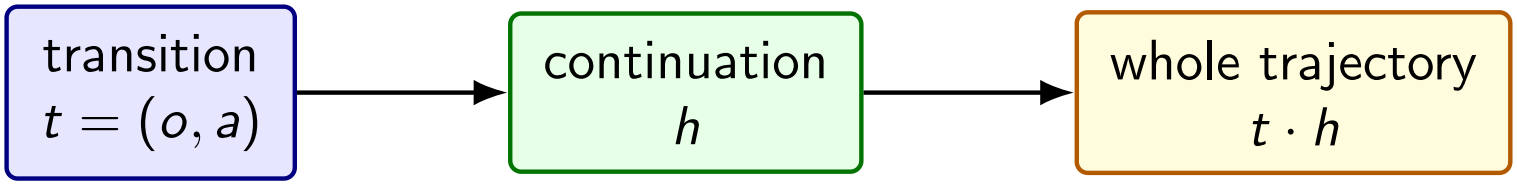
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- But the increment $r(t; h)$ may be non-local, and hence cannot be optimised for directly.

Axiom 5: Temporal equanimity

Let $T = \mathcal{O} \times \mathcal{A}$ be the transition set.

- The fifth axiom says there is a function $\gamma : T \to [0, 1]$ such that prepending t rescales continuation differences, so that

$$u(t \cdot \mu) - u(t \cdot \nu) = \gamma(t)\big(u(\mu) - u(\nu)\big).$$

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- $0 < \gamma(t) < 1$: future differences are damped.
- $\gamma(t) = 0$: after t, continuation does not matter.

This is transition-dependent discounting, not necessarily a single global constant.

Theorem 2: Markov reward representation

Assumptions: Axioms 1-5 over lotteries over trajectories.

- Then, utility has a local recursive form:

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This establishes a bridge from expected utility maximisation over whole-trajectory utility to RL-style expected sum of future reward maximisation.

Unrolling the recursion

- For $h = (t_1, t_2, \ldots, t_n)$:

$$\begin{aligned} u(h) = & \, r(t_1) + \gamma(t_1) r(t_2) \\ & + \gamma(t_1)\gamma(t_2) r(t_3) + \cdots \\ & + \left(\prod_{i=1}^{n-1} \gamma(t_i) \right) r(t_n). \end{aligned}$$

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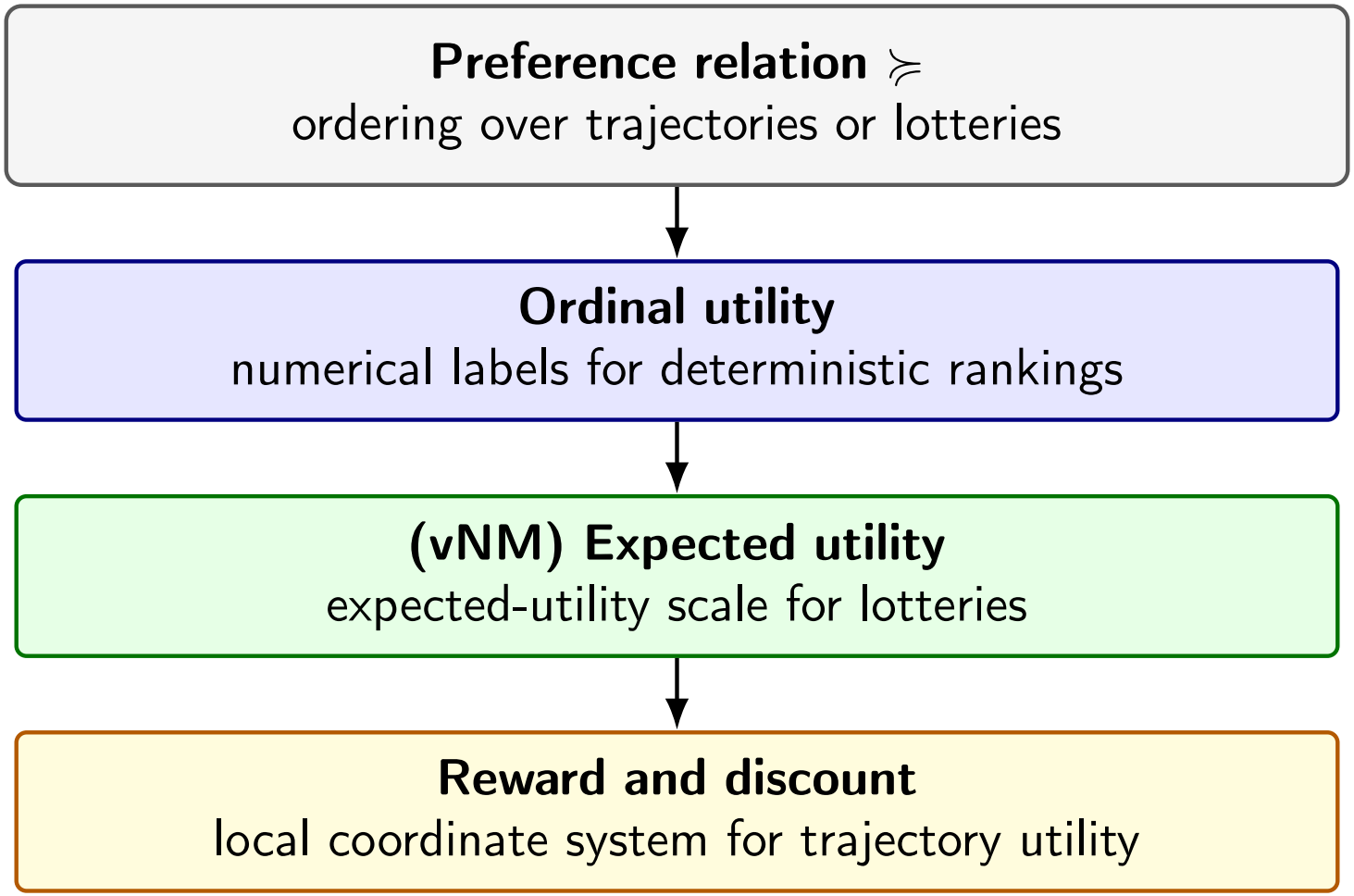
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- If $\gamma(t) \equiv \gamma$, this becomes standard discounted return:

$$u(h) = \sum_{k=1}^{n} \gamma^{k-1} r(t_k).$$

Summary: Preference, utility, and reward



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  - Robust values? (e.g. continuity)

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- Instead of 'given a rewards, how to optimise it' we could ask 'what kind of properties we want our reward to satisfy'. For example, Are any of these properties (or their negation) useful for fostering:
  - Corregibility? (e.g. incompleteness)
  - Robust values? (e.g. continuity)
  - Something else?
- Is RLHF providing an ordinal or cardinal utility signal? Are there implications of this?

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